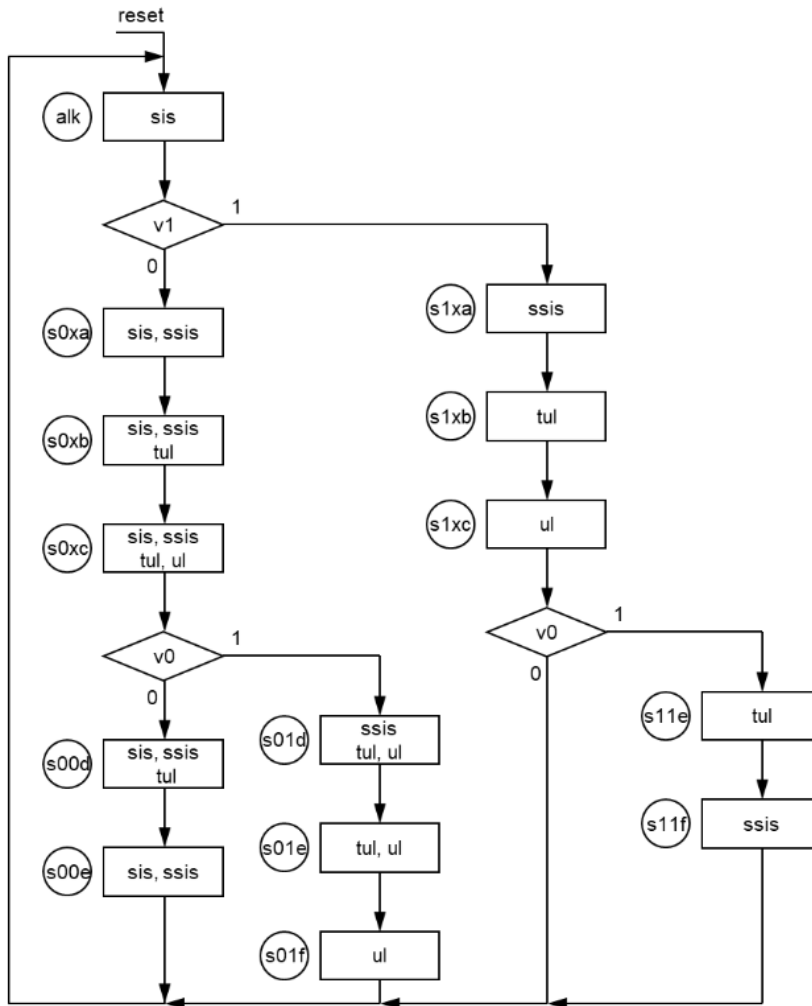


1.1

- What is the minimum number of flip-flops (state bits) needed for the following ASM-chart?
- How many output signals does the system have?
- Is it a Mealy or Moore state machine?

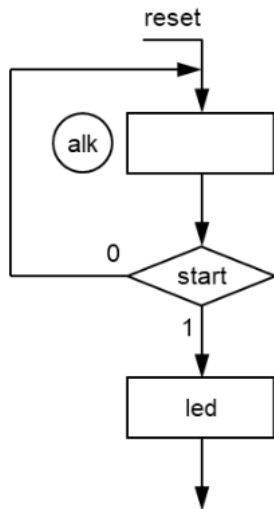


There are 14 states so 4 flip flops are needed, this can be proven via $\log_2(14)$ if needed.

Outputs are sis, ssis, tul and ul, for a total of 4.

Moore, since there are no states with round corners which would be used to represent a mealy-type state machine.

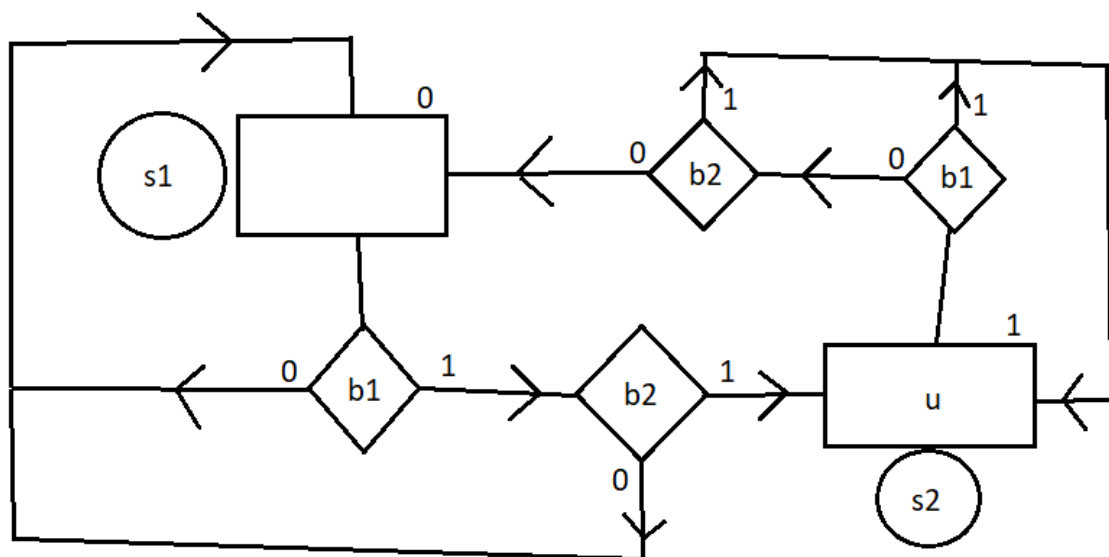
- 1.2 In the following ASM-chart the clock cycle time is 0.5 s. How many states do you need to add if motor should be started 2 seconds after the button was pressed?



Two states should be added between 'start' and the motor (which I assume is started one clock tick after the last depicted state as that bottom arrow implies) so that the expected wait time is $\sim 1,51-2$ s depending on when in the clock cycle the button is pressed. If it is imperative that the machine waits **at least** two seconds. Then a fourth state should be added to the chain to increase the wait time to $\sim 2,01-2,5$ s.

- 1.3 A bank vault door is controlled by a sequential circuit. Initially the door is locked. The door can be opened by pressing two opening buttons simultaneously. The buttons are placed at opposite ends leading to the vault. When the buttons are pressed the unlock signal is activated. When the unlock signal is active a yellow warning light flashes. The unlock signal stays active until both buttons are released. When both buttons are released the unlock signal is deactivated, flashing stops, and the circuit returns to the initial state.

Name signals and draw an ASM-chart of the door controller.



The 'u' output is the unlock signal, 'b1' is the first button, and 'b2' is the second button.